

$$\text{ex } a^0 = 1$$

⇒ Define the determinants of a square matrix

$$\text{ex- } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

ex)

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} +$$

$$a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\text{ex 1. } \begin{vmatrix} 3 & 7 \\ 1 & -4 \end{vmatrix} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\text{ex 2. if } A = \begin{vmatrix} 4 & -3 & 5 \\ 1 & 0 & -3 \\ -1 & 5 & 2 \end{vmatrix}$$